[INCH-POUND]
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FEDERAL SPECIFICATION

MIXER, CONCRETE, TRUCK-MOUNTED

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a truck-mounted, diesel-engine driven, inclined axis revolving drum-type mixer. The truck mixer shall have a capacity of not less than 7.5 cubic yards (5.73 cubic metre (m^3)) of mixed concrete when used to mix, transport, and place concrete. This specification covers only one type of mixer.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Specifications

MIL-V-173	-	Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-H-13531	-	Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 1581), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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Military Standards

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223 - Nontactical Wheeled Vehicles; Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.
MS 51118 - Pintle Assembly, Towing: 40,000 Pound Capacity, Manual Release.

(Copies of federal and military specifications, and standards required by contractors in connection with specific procurement functions are obtained from the Defense Automated Printing Services, Attn: DoDSSP, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.1 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

(Copies of specifications, standards, handbooks, drawings and publications required by manufactures in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings, Standard.
SAE J537 - Storage Batteries, Standard.

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SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic radiation from Vehicles and Devices (30 to 1000 Mhz), Standard.
SAE J588 - Turn Signal Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width, Standard.
SAE J589 - Turn Signal Switch, Standard.
SAE J682 - Rear Wheel Splash and Stone Throw Protection, Recommended Practice.
SAE J688 - Truck Ability Prediction Procedure, Recommended Practice.
SAE J704 - Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs, Standard.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Copley, OH 44321.)

TRUCK MIXER MANUFACTURERS BUREAU (TMMB)

Truck Mixer and Agitator Standards.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Truck Mixer Manufacturers Bureau, 900 Spring Street, Silver Spring, MD 20910.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The mixing unit shall consist essentially of a frame, rotating drum, charging hopper, discharge chute, and a water system. The mixing unit shall be mounted on a diesel engine driven, six by six, or six by four truck chassis. As used herein, the term “mixer” means the mixing

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unit only. The term “truck mixer” means the truck chassis and mixer. The term “truck” is used to describe only the truck chassis.

3.1.1 Standard commercial product. The truck mixer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer’s standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer’s standard commercial product, shall be included in the truck mixer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer’s catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish one truck mixer for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Prohibited materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations (CFR), Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Maintainability. The truck mixer shall be designed so that accessories and components are readily accessible for repair or replacement with minimum removal or disturbance of adjacent parts or components, using general purpose tools. Operating parts, accessories, and drain outlets shall be readily accessible for regular maintenance service. Drainage of all liquids shall be by gravity onto the ground. Integral tubes or troughs may be used to convey the liquids from the drain outlet to the ground.

3.6 Safety. All moving parts which are of such a nature or so located as to be a hazard to operating personnel shall be fully enclosed or guarded. Projectile devices shall not impair the operating functions of the truck mixer. Platforms, steps, and rails shall be furnished, and shall be located where necessary to ensure safe, easy access by personnel to those areas requiring frequent

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maintenance or service. All walking surfaces, except those decked with open pattern steel plate, shall be painted with a nonskid, abrasive type, deck paint. Truck mixer shall be provided with the following equipment:

a. Fenders. Full length rear fenders equipped for splash and stone throw protection as specified in 3.29.

b. Signaling devices. An audible, pulsating, signaling device to caution personnel when truck is in reverse gear operation.

c. Platform. An operator's platform located near the charging hopper.

d. Steps. Either steps or a ladder of sufficient length to ensure safe and easy access, by the operator, from ground surface to charging hopper platform.

e. Ladder. When truck mixer is equipped with a pedestal mounted water tank, a ladder shall be provided to allow operator safe and easy access from the ground surface to top of water tank. If easy and safe access to the tank fill point is available from the truck access step and handle, this will be acceptable.

3.7 Mixer performance. Mixing and agitating speeds, measured in revolutions per minute (rpm), shall conform to the TMMB Truck Mixer and Agitator Standards. With all ingredients charged into the drum and operating at mixing speed, the mixer shall thoroughly mix the ingredients into a homogeneous mixture within the number of revolutions specified by the TMMB Truck Mixer and Agitator Standards. The truck mixer shall be capable of mixing and agitating a full load of concrete while in transit and at an average travel speed of 35 miles per hour (mph) (56 kilometre per hour (km/h).

3.8 Drum. Unless otherwise specified (see 6.2), the drum shall be the open end type. Capacity and volume of the drum shall conform to table I. The basis for determining drum capacity and the formulas used for calculating drum volume shall be as specified in the TMMB Truck Mixer and Agitator Standards. Drums of larger nominal capacity will be acceptable provided maximum and minimum gross volumes conform to the TMMB Truck Mixer and Agitator Standards. The drum shall be watertight and made of corrosion-resistant and abrasion-resistant steel. All sections shall be of welded construction to form a smooth uniform surface enabling the entire drum to withstand the maximum loads or stresses during heavy operations. The drum shall be equipped with an adequate number of blades to thoroughly mix the concrete and to quickly discharge the homogeneous batch. The blades shall be of the same material as the drum. The blades shall be either heat treated or their leading edges shall be either heat treated or hard surface welded. Drum shall be equipped with quick opening type inspection hatch(es). When hatch(es) are closed, drum shall be watertight. Drum rollers shall be made of high strength material, precision machined, and treated to resist wear.

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TABLE I. Capacities and volumes (in accordance with TMMB).

Capacity, cubic yard mixed concrete		Gross Volume, cubic feet inclined centered axis
As Mixer	As Agitator	Minimum
7.5	9	329
(5.7 m^3)	(6.9 m^3)	(9.3 m^3)

3.9 Water system. Mixer shall be equipped with an air-pressurized water system or a mix and flush water tank(s) system. The water tanks(s), automatic water meter control, and water system shall comply with the requirements of the TMMB Truck Mixer and Agitator Standards, except water tank(s) shall be not less than 200 gallons (757 litre (L)). Sight gages shall be visible to the operator while standing at the rear controls. Water pump, when provided shall be of the centrifugal type, in the heavy-duty class, and shall be mechanically or hydraulically driven. Mixer shall be provided with a dual water injection system, a 25-foot (635 mm) washoff hose with adjustable spray and shutoff nozzle, and drain cocks to completely drain the water system. The mixer, when equipped with a mix and flush tank(s) system shall utilize a water recirculating system to prevent freezing of water in the system during cold weather operation. When equipped with an air-pressurized water system, an air blowout system shall be used to drain the water. Plastic piping will not be acceptable. The water meter furnished shall be resettable counter type, with totalizer and automatic stop valve.

3.10 Charging hopper. Unless otherwise specified (see 6.2), the manufacturer shall furnish his standard commercial stationary or swing out type charging hopper.

3.11 Discharge chute. Discharge chute shall consist of an upper and a lower section. The discharge end of the chute shall be horizontally and vertically adjusted. Upper section shall be pivotally attached to mixer frame to provide not less than 180 degrees horizontal movement of chute end. Vertical adjustment of chute end shall be accomplished through the use of a hydraulic cylinder attached to the upper chute section. Upper and lower sections shall be hinge connected. Hinge connection shall allow lower section to fold over upper section for transit stowage. An extension chute section(s) shall be furnished. With chute extension section(s) attached to upper and lower chute section, overall length of the chute shall be not less than 12 feet (3 657.6 mm). Chute extension section(s) shall be constructed to allow quick and easy connection and disconnection. Stowage space shall be provided on truck mixer for securely stowing disconnected chute extension(s). Discharge chute with extension(s) attached shall have a smooth, unobstructed trough.

3.12 Power transmission. A hydrostatic transmission shall be furnished and unless otherwise specified (see 6.2), either open or closed circuit type will be acceptable. The hydraulic pump shall be mounted on, and driven by, the truck's power takeoff as specified in 3.23. The hydrostatic shall be complete with all operating accessories, including the pump, motor(s), reservoir, valves, filters, hoses, piping, fittings, and when applicable, a heat exchanger and temperature gage(s). All hoses and fittings shall be capable of withstanding a bursting pressure of four times the working pressure and a proof pressure of not less than two times the working pressure. Hoses shall have

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physical qualities equal to or conforming to MIL-H-13531, type I or II, as applicable. Hoses shall be protected against chafing through utilization of supports and protective devices. The hydrostatic transmission shall be equipped with a full flow filter on the pressure side of system. The inlet side shall have a suction strainer or filter to prevent entrance of foreign material. Means shall be provided to vent the reservoir. The vent shall be equipped with a micronic filter element to prevent dirt from entering the reservoir. Hydraulic oil shall be of the noncorrosive type with the characteristics to withstand a temperature range 10 degrees Fahrenheit (°F) to 325 °F (-12 degrees Celsius (°C) to 162.8 °C). Reducer sprocket, shafts, and chains shall be heat-treated alloy steel. Provisions shall be made for adjusting chain tension. An automatically operated oiler shall be furnished to lubricate the chain. Hydraulic motor or direct gear drive is acceptable.

3.13 Revolution counter. An electrically operated, limit switch type, or a mechanical type drum revolution counter shall be furnished.

3.14 Controls. Truck mixer shall be equipped with two sets of mixing drum controls. Each set of controls shall control engagement and disengagement of the power train, rotary speed, and direction of rotation. One set of controls shall be located in the truck's cab; the second set shall be located at or near the left rear corner of the mixer. Cab controls shall be within easy reach of the operator while sitting in a normal driving position. Rear controls shall be within easy reach of the operator while standing on the ground. All controls shall be clearly identified for their intended function with permanently affixed identification plates made of corrosion-resistant material.

3.15 Mixer frame. Mixer frame shall support mixer weight at pedestals, and will mount on the truck chassis frame as specified in 3.24. When rivet or bolt heads protrude above the truck chassis rails, metal sills shall be installed between the mixer frame and chassis rails. Counterbored holes shall be drilled into the sills to receive the corresponding rivet or boltheads.

3.16 Truck chassis. Mixer shall be mounted on a diesel-engine-driven six-wheel, four-wheel drive, truck-chassis having a minimum gross vehicle weight (GVW) rating of not less than 62,000 pounds (lb) (28 117 kg). Except as otherwise specified herein, all chassis items shall be as represented in the chassis manufacturer's technical data and the mixer shall be as represented in the mixer manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of truck models and mixers, and shall be on file in the engineering offices of the procuring activity prior to delivery of the truck mixer. The truck mixer shall comply with all DoT Federal Motor Vehicle Safety Standards and Regulations applicable on the date of manufacture.

3.17 Vehicle weight and rating.

3.17.1 Curb weight. Curb shall include weight of chassis and cab, all attachments, accessories, equipment, and full complement of fuel, lubricants, and coolant.

3.17.2 Weight rating. The GVW shall consist of the curb weight, a 175-lb (79.4 kg) operator, and a payload to provide a weight rating not less than 62,000 lb (28 117 kg).

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3.17.3 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the vehicle loaded to its maximum mixer capacity. Suspension, axle, and tire rating shall be not less than equal to the load imposed, measured at the ground.

3.18 Dimensions. Overall width of truck mixer shall be not greater than 96 inches (2 438.4 mm). The truck mixer shall be capable of being loaded on a Landing Ship Tank by being driven or towed on board when the ramp is in the fully lowered position as shown in figure 1.

3.19 Truck performance. High and low speed requirements shall be met with truck loaded to specified GVW. The truck shall ascend a 0.50 percent grade at 44 mph (70.8 km/h) when calculated in accordance with SAE J688 (see 6.4). With drum filled to mixing capacity and operating at not less than 2 rpm and not more than 6 rpm, truck mixer shall be capable of maintaining a constant speed of 35 mph (56.3 km/h) on a relatively level improved road (see 6.5.1). The low speed requirements shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall be not greater than 3 mph (14.5 km/h). With drum filled to agitating capacity and operating at not less than 3 mph (14.5 km/h), truck mixer shall be capable of traveling on unimproved roads (see 6.5.2).

3.20 Engine. Truck shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. The engine shall have horsepower, torque, and speed characteristics to meet satisfactorily or exceed all vehicular performance requirements specified herein. Engine shall be equipped with a full flow type oil filter. An engine governor shall be furnished, set, and sealed to limit the maximum engine operating speed to the recommended value. Engine shall comply with EPA regulations governing Control of Air Pollution from New Motor Vehicle and New Motor Vehicle Engines applicable on the date of manufacture. Vehicles with California destinations shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.20.1 Engine starting requirements. The diesel engine shall start in any temperature above 0 °F (-17.8 °C). Starting aids may be either electric glow plug or ether primer. When an ether priming system is required, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces (355 millilitre (mL)).

3.21 Truck systems.

3.21.1 Electrical. Truck shall be equipped with a 12-volt direct current starting and lighting system, with not less than a 50-ampere generator or alternator. All lights, reflectors, and wiring shall be as specified herein and shall be in conformance to DoT Motor Carrier Safety Regulations. Turn signal units conforming to SAE J588, class A, with SAE J589, class A, operating units mounted on the steering column, shall be furnished. Truck shall be provided with double faced front signal units and with single faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall be self-canceling type with visible and audible flash indicator.

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3.21.1.1 Dry charged batteries. Batteries furnished shall be dry charged in accordance with SAE J537, without electrolyte, with sealed caps to prevent intrusion of atmospheric moisture.

3.21.2 Fuel. Fuel system shall conform to DoT Motor Carrier Safety Regulations, section 393.65. Fuel tank(s) capacity shall be not less than 100 gallons (378.5 L). When more than one tank is furnished, means shall be provided to ensure equalized fuel level in both tanks. Fuel filters shall be installed at the tank and before the engine. The intake air cleaner shall be a two-stage, dry-type, equipped with a service indicator.

3.21.3 Cooling. The truck shall be equipped with a heavy-duty cooling system. The system shall maintain coolant at a temperature below the boiling point when the truck is loaded to rated GVW and operating at 10,000 feet (36 480 mm) above sea level, or in an ambient air temperature of not less than 125 °F (51.7 °C).

3.21.4 Sound level. Noise levels shall be less than 85 dB(A) at the operator's position and other occasionally occupied positions as measured during typical operating conditions and environments. If noise levels cannot be reduced to below 85 dB(A), then hearing protection warnings shall be placed on the unit where they can be read by those personnel exposed to the hazard.

3.22 Power train. Truck shall be equipped with a transmission having not less than 10 forward speeds. Transmission shall be furnished with two power takeoff openings conforming to SAE J704. Axles and transmission gear ratios shall be matched to provide a progressive shifting pattern throughout the complete range. Transmission shall be easily shifted in all speeds. Input torque capacity of transmission shall be not less than equal to the maximum torque delivered by the engine. Drive line components shall transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction, without sustaining any damage or deformation. Truck shall be equipped with the largest capacity clutch offered for its size. Torque capacity of the clutch shall be greater than the maximum delivered engine torque.

3.23 Power takeoff. Engine shall be furnished with an independent type power takeoff to drive the hydraulic pump specified in 3.12. Power takeoff shall be located either at the flywheel or front end of the crankshaft. When a front end power takeoff is furnished, the underside of the hydraulic pump, piping, and hoses shall be guarded to protect against damage from obstacles projecting from ground. Guard shall be constructed of steel plate having a thickness of not less than 0.25-inch (6.4 mm). The channel type front bumper specified in 3.28 shall be reinforced and positioned to the front of the hydraulic pump.

3.24 Frame, suspension, and axles. The truck shall be furnished with a reinforced standard frame. Reinforcement shall provide sufficient structural strength in the chassis frame through increased resistant bending movement to not less than equal the loads imposed, with the truck loaded to provide specified GVW. The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. Bogie shall be provided with a means permitting differential action between the two axles, and a manually or automatically controlled lockout ensuring equal power to each rear axle. When a manually controlled lockout is furnished,

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lockout controls shall be located in the truck cab. Axle gear ratios shall provide performance as specified in 3.19.

3.25 Wheels, rims, tires, and tubes. Unless otherwise specified (see 6.2), the truck shall be equipped with single wide base wheels and tires on the front and rear. Centerline of front and rear wheels shall be equidistant. Rim and tire rating shall conform to TRA recommendations for the type and size tires furnished. Tires shall be of the tube or tubeless type with mud and snow tread. Tires shall be of rated capacity not less than equal to that specified in 3.17.3. Tires shall be of not less than 100 level quality. Inner tubes shall be of the heavy-duty type and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations. Truck shall be furnished with a spare wheel and tire, and mounted on a carrier.

3.26 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations, sections 393.45 through 393.52, as applicable. Service brakes and parking brakes shall conform to Federal Motor Vehicle Safety Standard No. 121.

3.26.1 Service brakes. The truck shall be equipped with full air brakes. The braking system shall be complete with all necessary components as follows:

a. Air compressor, unloader-head type, engine driven and lubricated, air or water cooled, having a capacity of not less than 112 cubic feet per minute (52.8 L/s).

b. Air storage reservoir(s) with not less than 3,000 cubic inch (1 835.3 cm^3) total capacity; each tank equipped with a drain and with safety and check valves between the compressor and the last reservoir tank.

c. Foot control, suspended or treadle type.

d. Air control valves.

e. Air pressure gage, visible to the driver.

f. Low air pressure warning, visible and audible.

g. Service brake stop light switch.

h. Alcohol aspirator with unbreakable transparent container.

i. Automatic moisture ejector.

3.26.2 Parking brake. Truck shall be equipped with a parking brake control valve. Parking brake shall conform to Federal Motor Carrier Safety Regulation 393.41.

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3.27 Cab. The truck shall be furnished with a standard full width cab not less than 60 inches (1 524 mm) wide measured between door panels. Doors shall be equipped with locks operable from inside the cab with both doors fitted with a key operated external lock. Cab shall have a full width seat and back, or individual driver and passenger seats and backs. In either case, seat(s) and backs(s) shall be upholstered. Driver's seat shall be adjustable. Safety grips shall be provided on each side of cab to assist personnel to enter or leave cab. Truck shall be furnished with dual windshield wipers and washers. Windshield wipers shall be multispeed type and operated by air or electric motor(s). A hot water personnel heater with blower and fresh air intakes shall be provided. Heater discharge outlets shall direct heated air to floor and defroster louvers. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be easily operable by and conveniently located to the driver. Lever controls shall be designed and located in the cab to permit easy entrance and exit of the driver. Instruments and controls shall be identified as to their function and installed to facilitate easy removal and servicing. The following equipment shall be furnished:

a. Ammeter or charging indicator.

b. Fuel gage.

c. Oil pressure gage or indicator.

d. Engine temperature gage or indicator.

e. Speedometer with recording odometer.

f. Tachometer.

g. Dual sun visors.

h. Armrest.

i. Cab ventilator other than window.

j. Airhorn.

k. Interior lights.

l. Two pairs of seat belts.

m. Power steering.

n. Outside rearview mirrors, mounted on each side of cab, each with not less than 85 square inches (54 838.6 mm^2) of reflective surface, and each with not less than three supporting arms.

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o. All other equipment usually furnished as standard by the manufacturer on his commercial vehicles.

3.28 Bumper and towing devices. A channel type front bumper shall be furnished. Not less than two hooks, loops, or pins for towing the truck mixer shall be furnished on front of truck, towing devices shall transfer forces directly to chassis' frame rails. When a front end power takeoff is furnished, the front bumper shall be reinforced as specified in 3.23. A pintle assembly conforming to MS 51118 shall be mounted at rear of truck. Mounting shall include reinforcement to transfer pintle loads directly to chassis frame rails.

3.29 Rear wheel splash and stone throw protection. Rear wheels shall be provided with ridged splash shields in front and rubber mud flaps at rear. Splash shield shall be constructed of metal or rubber with metal trim. Splash and stone throw protection shall be in accordance with SAE J682.

3.30 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (6 895 kilopascals) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.31 Lifting and tiedown attachments. When specified (see 6.2), the truck shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the truck. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tying down attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the truck on the carrier when shipped.

3.32 Electromagnetic interference characteristics. When specified (see 6.2), truck mixer shall conform to the electromagnetic characteristic requirements and test limits of SAE J551.

3.33 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.

b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.34 Plates.

3.34.1 Identification plates. An identification plate will be furnished by the contracting officer for each truck. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each truck in a conspicuous place with nonferrous screws, rivets, or bolts not less than 0.125-inch (3.2 mm) in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.34.2 Instruction plates. The truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, rivets, or bolts of not less than 0.125-inch (3.2 mm) diameter.

3.34.3 Rating. A certified standard rating plate furnished by the TMMB shall be permanently attached to the truck mixer.

3.34.4 Data. The mixer manufacturer shall furnish a data plate conforming to requirements of the TMMB Truck Mixer and Agitator Standards.

3.35 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substance. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surface shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.063 5 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.36 Workmanship.

3.36.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.36.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.36.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.36.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.36.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.37 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the truck mixer when a first article is required (see 3.2, 6.2. and 6.3). This inspection shall include the examination of 4.3 and the tests of 4.4.1 through 4.4.7. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the test of 4.4.8.

4.3 Examination. Each truck mixer shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Weight tests. Truck mixer shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. To verify that the distribution of GVW is not exceeded, as specified in 3.17.3, the imposed loading on front and rear axle shall be computed using the curb weight, the operator weight at 175 lb (79.3 kg), and the mixer payload.

4.4.2 Water calibration test. Water tank(s) capacity shall be determined to ascertain compliance with the requirements of 3.9. To verify conformance to the requirements of the TMMB Truck Mixer and Agitator Standards, the automatic water meter shall be tested for accuracy and the water pump when furnished, for discharge capacity.

4.4.3 Road test. Truck mixer with payload shall be examined and road tested by the contractor to confirm that the truck mixer will operate in accordance with contract requirements. The Government reserves the right to perform inspection requirements on the first article truck mixer with payload, where such inspections are deemed necessary to ensure the truck mixer will operate in accordance with the contract requirements. When applicable, road test may be performed in conjunction with the agitating and mixing tests.

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4.4.4 Mixing test. Water tank(s) shall be filled to capacity. Mixing drum shall be charged to maximum rated mixing capacity with the following materials measured by weight: one part cement, three parts sand, and five parts coarse aggregate. Seven gallons (26 L) of water, from the truck mixer water tank(s), shall be added for every 95 lb (43 kg) bag of cement used. After the drum has been charged, the truck mixer shall be driven at varying speeds up to 35 mph (56.3 km/h) for not less than 10 miles (16.1 km) on an improved road. While in transit, the drum shall be operated at not less than 2 rpm, nor greater than 6 rpm. The amount of mixing shall be not less than 70, nor more than 100 revolutions which shall be verified with the revolution counter. The batch shall be discharged and examined to determine that the batch is thoroughly mixed and all particles of aggregate are coated with cement slurry. Mixer shall discharge clean, leaving no collection of concrete inside the drum.

4.4.5 Agitating test. Water tank(s) shall be filled to capacity. Drum shall be charged with not less than 7.5 cubic yards (5.7 m^3) of wet concrete. After drum has been charged, the truck mixer shall be driven at varying speeds for not less than 10 miles (16.1 km) on unimproved roads. Truck mixer shall make not less than 10 stops. During 5 of the 10 stops, the truck mixer shall reverse its direction of travel by backing up a distance of not less than 50 feet (15 240 mm). While in transit, the drum shall be operated at not less than 2 rpm and not more than 6 rpm. At the completion of the transit part of this test, the batch shall be discharged and examined to determine that the batch is thoroughly mixed and all particles of aggregate are coated with cement slurry. Throughout the test, the truck mixer shall be observed to ascertain that no appreciable amount of concrete is lost due to spillage.

4.4.6 Lifting and tying down attachment test. When lifting and tying down attachments are required, they shall be tested to determine conformance to 3.31.

4.4.7 Test for electromagnetic interference characteristics. When electromagnetic interference characteristics are required, the truck mixer shall be tested to determine conformance to 3.32. In lieu of test to determine conformance to SAE J551, the manufacturer may furnish a certification that the truck mixer meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.4.8 Production unit test. Each truck mixer shall be completely assembled, properly adjusted, and otherwise serviced for operation. Each truck mixer shall be operated for not less that 1 hour with the drum rotating not more than 12 rpm and not less than 4 rpm. During this part of the test, truck mixer shall be driven not less than 5 miles (8 km) on improved road and 5 miles (8 km) on unimproved roads. The following shall be tested for compliance with specified requirements:

a. Inspection hatches (see 3.8).

b. Water pump discharge capacity (see 3.9).

c. Automatic water meter (see 3.9).

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d. Discharge chute (see 3.11).

e. Revolution counter (see 3.13).

All controls shall be operated as many times necessary, but not less than 10 times, to determine ease of operation, effectiveness and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gearcases, bearing, shafts, chain drives, and other moving parts shall be visually examined for excessive heating or abnormal operation insofar as is practicable and possible. Truck mixer components shall be examined for evidence of failure, slippage, or deformation. Failure to pass any phase of this test shall be cause for rejection.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The concrete mixing truck is intended to be used to mix, transport, and place concrete at construction sites.

6.2 Acquisition data. Acquisition documents should specify the following:

a. Title, number, and data of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
c. When closed end drum is required (see 3.8).
d. When other than stationary or swing out type hopper is required (see 3.10).
e. When open or closed circuit type hydrostatic transmission is required (see 3.12).
f. When different type tires are required (see 3.25).
g. When lubrication is other than specified (see 3.30).
h. When lifting and tiedown attachments are required (see 3.3.1).
i. When conformance to electromagnetic interference characteristic requirements and test limits are required (see 3.32).
j. When fungus resistance is required (see 3.33).
k. Color of finish coat (see 3.35).
l. When rustproofing is required (see 3.37).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one truck mixer. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirements for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheel-base, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.19) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	-	Tire Factor. This factor relates to the size of tires furnished by the contractor in accordance with this specification.	
Table 2	-	Altitude Factor.	1.00
Table 3	-	Rolling Factor.	1.356
Table 4	-	Area Factor.	0.173
Table 5	-	Velocity Factor.	170.0
Table 6	-	Altitude Factor.	1.0
Table 7	-	Chassis Fraction Horsepower. Use unit GVW to nearest, higher 1,000 lb (454 kg) and engine rpm to nearest 100 revolutions which is required for 44 mph (71 km/h) geared speed. for GVW and engine speed beyond the range of this table, factors shall be extrapolated.	
Table 8	-	Grade Factor.	0.85
Table 8A	-	Correction Factor.	Not required
Table 9	-	Road Factor.	0.0

6.5 Definitions.

6.5.1 Improved road. An improved road has a smooth, hard surface such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road has an unstabilized base and an undulating, unpaved surface, having occasional chuckholes and exposed rocks.

6.6 Supersession data. This specification replaces Military Specification MIL-M-911G(YD) dated 7 February 1992.

6.7 Subject term (key word) listing.

Cab
Chassis
Engine
Hopper
Frame
Wheels

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6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the superseded document.

MILITARY INTEREST:

Custodians:
Navy - YD1
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Navy - YD1

(Project 3895-0036)

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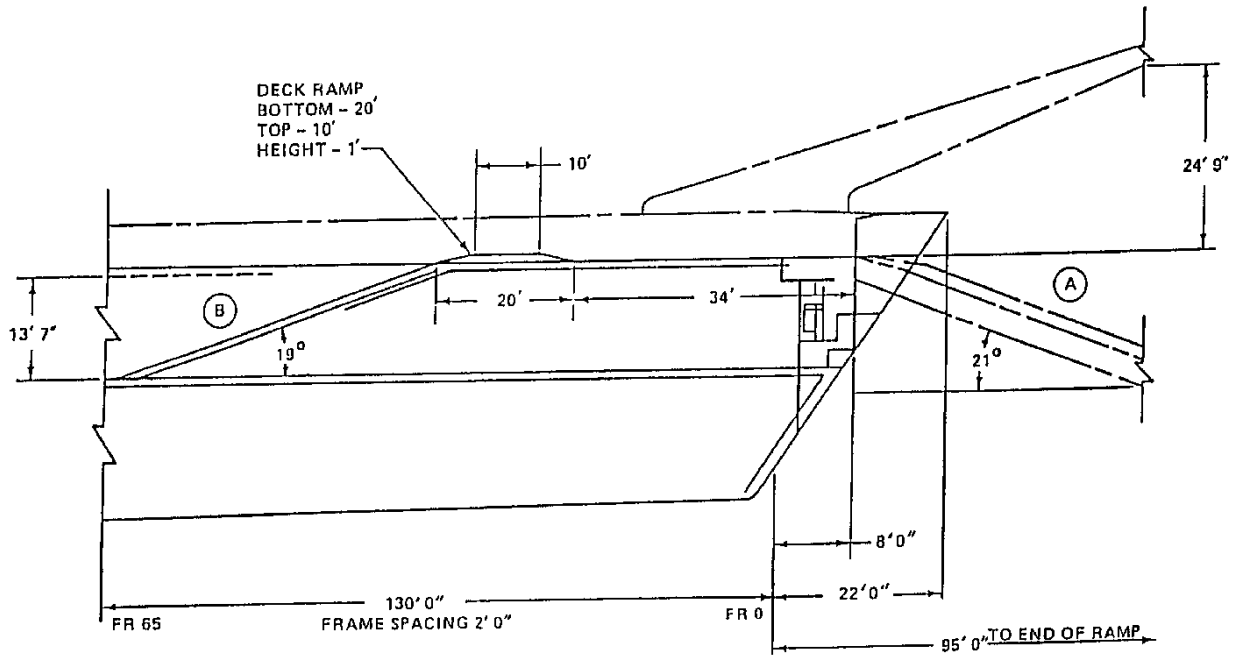


FIGURE 1 Ramp.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** OO-M-2897	**2. DOCUMENT DATE (YYMMDD)** 970314

3. DOCUMENT TITLE
MIXER, CONCRETE, TRUCK MOUNTED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME RUDY PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial: 805-982-5843 (2) AUTOVON: 551-5843
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2D 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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